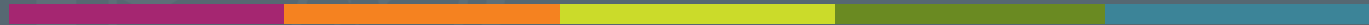


The Next Era of Assessment – Innovating for the Future



Mohammed Minhaj, MD, MBA
Director, American Board of
Anesthesiology

David A. Turner, MD
Vice President, Innovation and
Competency-Based Education
American Board of Pediatrics

Cameron Wright, MD
Executive Director, American Board of
Thoracic Surgery

ABMS Conference
September 2026



Overview/Objectives



Describe

the concept of innovation and the importance of innovation for training programs and ABMS member boards

David A. Turner, MD



Discuss

the value of a trainee engagement strategy and value of a longitudinal assessment program for learners

Mohammed Minhaj, MD, MBA

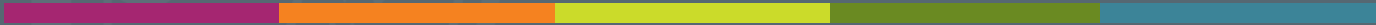


Explain

the challenges and benefits of using Artificial Intelligence to facilitate generation of items for certification assessments

Cameron Wright, MD

Assessment, Innovation, and Certification



David A. Turner, MD

Vice President, Innovation and Competency-Based Education

American Board of Pediatrics

ABMS Conference

® September 2026

A young woman with red hair and freckles, wearing a white lab coat over a blue collared shirt and a black tie. She has clear safety goggles perched on her head. She is looking directly at the camera with a slight smile. In the background, several other students in white lab coats are blurred, suggesting a classroom or laboratory setting.

No disclosures

The background of the slide is a close-up photograph of a person's face, focusing on the eye and nose. A camera lens is held in the foreground, partially obscuring the face. The lens reflects a bright yellow flower, likely a daisy, which is in sharp focus. The overall lighting is warm and soft, with a blue semi-transparent rectangle overlaid on the center of the image to contain the text.

OUR VISION

Advancing the
health of all children.

What is Innovation, and how does it fit into certification?



A group of healthcare professionals, including doctors and nurses, are seated in a circle on a patterned rug in a meeting room. They are holding clipboards and looking at each other, suggesting a collaborative discussion. The room has a chalkboard with various medical diagrams and posters on the wall, and a bookshelf with books and a potted plant. The text is overlaid in the center of the image.

Innovation is the deliberate evolution of certification to better ensure physicians remain fully prepared to meet the changing needs of patients, families, and society

Why innovate as a certification organization?

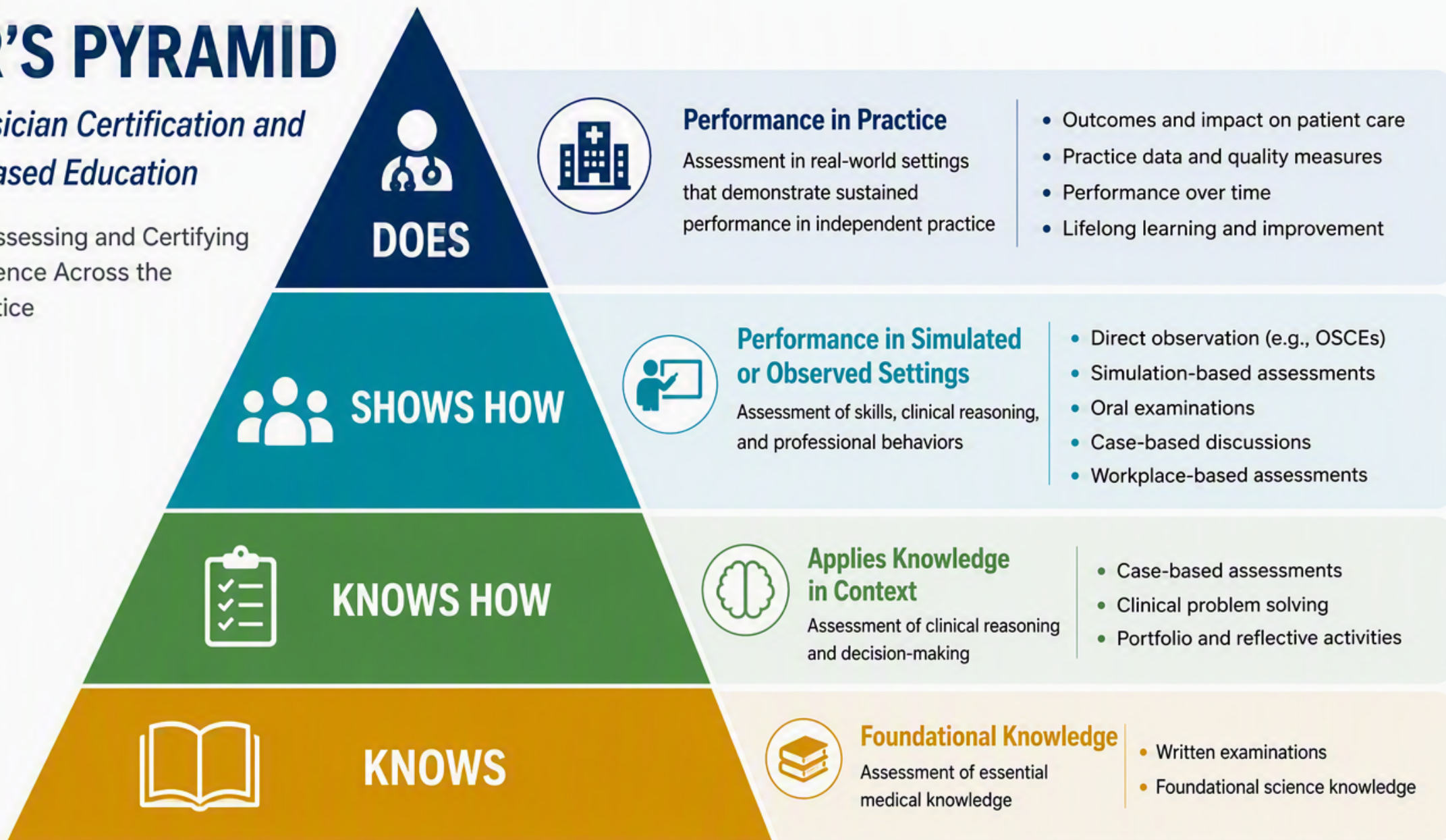
- Respond to a rapidly changing healthcare environment
- Maintain organizational relevance and trust
- Meet public expectations for physicians' competence
- Improve learning through better assessment approaches



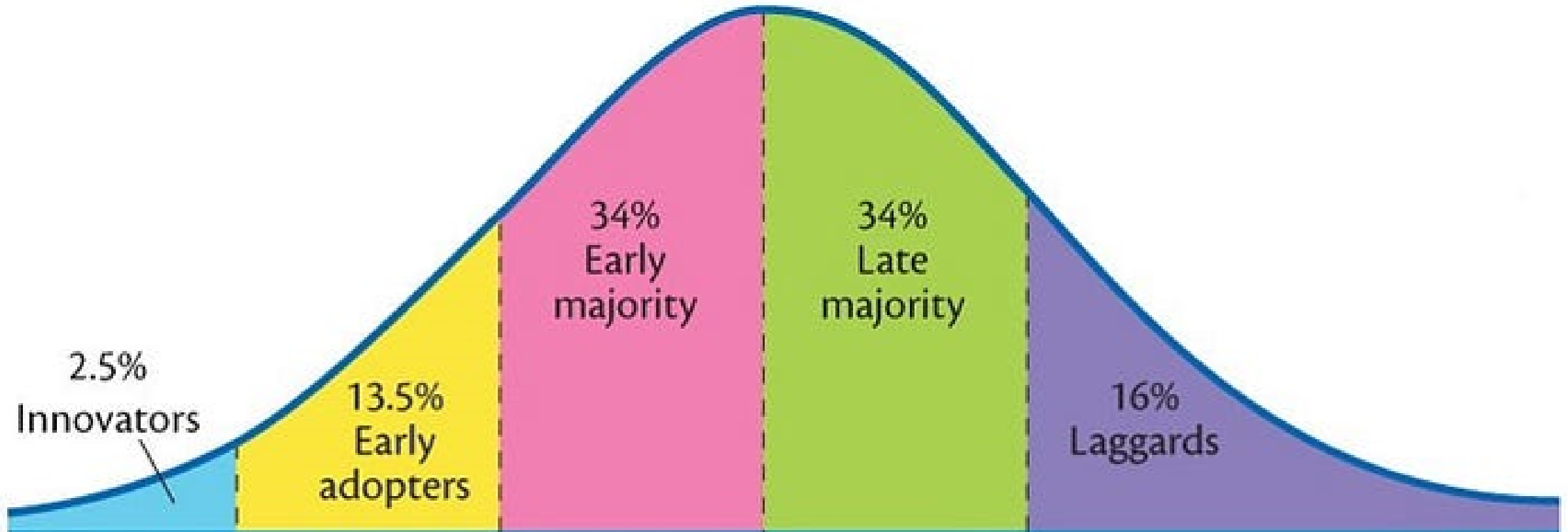
MILLER'S PYRAMID

Applied to Physician Certification and Competency-Based Education

A Framework for Assessing and Certifying Physician Competence Across the Continuum of Practice



Diffusion of Innovation - Many Layers



What does innovation look like at the American Board of Pediatrics?



Image created by Microsoft Copilot



ABP STRATEGIC PLAN
2025-2027

OUR STRATEGIC PRIORITIES:

Reshape Training and
Initial Certification through
Integration of CBME Principles

Transform Assessment
Development Process

Innovate and Expand
the Assessment of Core
Competencies

Strengthen Training Program
and Trainee Engagement

Drive Workplace Excellence

FOUNDATIONAL PRIORITY

Embed a commitment to diversity,
equity, inclusion, and belonging in
all the work of the ABP

THE AMERICAN BOARD *of* PEDIATRICS

Competency-based Medical Education

- By 2028 - EPAs (core outcomes of training) for initial certification
- Smartphone app to facilitate assessment workflow

Transform Assessment Development Processes

- Exploring AI for ABP's assessment processes

Innovate/Expand Assessment of Core Competencies

- Communications Assessment in Continuing Certification
- New Item type – Clinical Reasoning

Strengthen Training Program and Trainee Engagement

Drive Workplace Excellence



Innovation does not look the same
across ABMS member boards



“Journey of a thousand miles
begins with a single step”

Laozi

Thank You

dturner@abpeds.org

ABTS Exam Innovations

Key Features and AI assisted MCQ writing

Cameron Wright
Executive Director ABTS



Oral Exam Concerns-Mitigation by a Key Features Exam

Construct Underrepresentation

- The ability to present more case scenarios in the same amount of time

Construct Irrelevance

- The ability to provide an “Expectation of Bounded Disagreement”
- No “new examiners”, only new questions that can be vetted before inclusion
- Uniformity of the cases presented to candidates
- Individual candidate flexibility in time allocation to each question
- Uniformity in the data accompanying the cases presented to each candidate
- Removes raters and hence diminishes the opportunity for bias

Results in more reliable and hence more valid test scores*

*Bordage G, Page G. (2018). The key-features approach to assess clinical decisions: validity evidence to date. *Adv In Health Sci Educ*, 23(5):1005-36.

Key Features Exams

- They are linked to a very specific clinical problem
 - frequently much more specific than how our oral protocols are written
 - Each scenario should have 2-5 key features that are tested
- Each KF consists of three parts
 1. It is a **crucial** step in the resolution of a **specific** clinical problem
 2. Knowledge of this step **discriminates** between a passing and a failing candidate
 3. It includes **all** the correct responses and, if relevant, any responses that are critical fails

Vague answers like “appropriately manages intraoperative bleeding” won’t suffice
- The clinical problems and their associated KF’s will comprise a Blueprint of the construct being tested

- Key Features Exam: Item Construction and Grading
 - Working in small diverse groups of SMEs
 - Select clinical scenario (generated from our test blueprint or ABTS taxonomy).
 - List the 2-5 key features (essential/discriminatory) to the resolution of scenario.
 - Create specific case (a KF “Item”) that requires resolution of 2-3 of KF’s.
 - Create questions (short menu or write in) to address those 2-3 KF’s.
 - Grading is at the “item” level
 - Each KF item is worth 1 point total
 - A KF item that addresses 3 KF’s with 5 correct answers is **1 point.**
 - A KF item that addresses 2 KF’s with only 2 correct answers is worth **1 point.**

- Key Features Exam: Guidelines at Working Group

- In test blueprint, we focused on previous oral exam items with failures.
- Stem is longer to provide all pertinent information with minimal extraneous data.
- Examinee may not go back during the exam, so no “cueing” examinee.
- Decision making in surgical procedure/skills is difficult to write and score.
 - Institutional/individual variations.
 - List as many accepted approaches as possible.
- In creating the answer key, we may not achieve consensus.
 - Acknowledge alternative approaches may be effective and safe (lengthy discussion).
 - Achieving 80% consensus is “acceptable”.
 - Is there a “killer” or “critical fail” response? May discover these during scoring.
- Grading will be manual.
 - If a response is appropriate but not included in rubric, we will have adjudication process.
 - Committee of 3-5 examiners to review results of exam and adjudicate any issues.

How...

- Clinical Problems are identified from a test blueprint
 - Utilized our existing oral exam protocols
 - Identified those protocols with moderate fail rates
- Each Clinical problem will have 2-5 Key Features identified
- Case Scenarios are constructed for each clinical problem
 - Each Case Scenario focuses on only 2-3 relevant KF's
 - Therefore, some clinical problems may require more than 1 case scenario
- Questions are created for each case scenario to assess the candidates understanding of those KF's
 - Short menu (15-25 choices)
 - *Best for ordering tests during diagnosis and evaluation*
 - Write in
 - *Short (2-6 word) responses*
 - *Best for establishing a differential and treatment decisions*
- The final test is organized around the case scenarios

Test Blueprint

1. Lung Neoplasm
 - a. Early stage
 - b. Single station N2 disease
 - c. Multi-station N2 disease
 - d. Central tumor requiring sleeve
 - e. Central tumor requiring pneumonectomy
 - f. Etc
2. Lung Benign
 - a. Parapneumonic empyema
 - b. Lung Abscess
 - c. Etc...
3. Etc...
4. Etc...

Clinical Problem:
Single station N2
disease

Relevant Key Features

1. Approach to staging...
2. Approach to induction treatment...
3. Selection of surgical approach...
4. Addressing a positive margin...

Case Scenario #1

Scenario describes the initial visit immediately after a diagnosis of lung cancer is obtained.

Addresses KF #1 and #2

Case Scenario #2

Scenario described the patient after induction therapy is completed and progresses to the operating room.

Addresses KF #3 and #4

How...

- Writing KF questions: formats
 - Constructed response (write-in)
 - Pros: Recommended for testing diagnoses and management; useful where language is relatively uniform and response can be 2-3 words
 - Cons: Requires human scoring in most cases (NLP may allow computer scoring)
 - Selected response (short and long menu multiple choice)
 - Pros: Computer-scorable, recommended for testing history/exam findings, labs/imaging
 - Cons: Somewhat less discriminating around the cut score than short answer

A 68 year-old man with BMI 36 is found to have a cardiac murmur by his internist. A transthoracic echocardiogram shows aortic stenosis, and he is referred for evaluation. He denies shortness of breath with exertion and is on no medications. On examination, his heart rate is 76 bpm and BP 115/60 mmHg; he has a systolic crescendo-decrescendo murmur along the left sternal border and clear lung fields. Review of echocardiogram demonstrates a peak gradient of 65mm Hg and mean gradient 42 mmHg with a left ventricular ejection fraction of 65%.

What diagnostic test(s) would you order? List up to 3.

1. _____
2. _____
3. _____

The exercise test shows a decrease in systolic BP of 12 mmHg, what tests/evaluations would you obtain at this point? Select up to 3

- | | |
|------------------------------|-----------------------------------|
| 1. Arterial blood gas | 7. Dental evaluation |
| 2. Pulmonary Function Test | 8. Head CT scan |
| 3. Carotid duplex ultrasound | 9. Transesophageal echocardiogram |
| 4. Coronary angiography | 10. Abdominal ultrasound |
| 5. Vein mapping | 11. CT angiography |
| 6. Allen's Test | |

A 68 year-old man with BMI 36 with no regular health care is found to have a cardiac murmur by his internist. A transthoracic echocardiogram shows aortic stenosis and he is referred for evaluation. He denies shortness of breath with exertion and is on no medications. On examination, his heart rate is 76 bpm and BP 115/60 mmHg; he has a systolic crescendo-decrescendo murmur along the left sternal border and clear lung fields. Review of echocardiogram demonstrates a peak gradient of 65mm Hg and mean gradient 42 mmHg with a left ventricular ejection fraction of 65%.

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Key Feature being addressed:

Given an adult presenting with asymptomatic severe aortic stenosis (AS), surgery is not indicated unless there is a positive exercise test or rapid progression of the AS (IIA). For very severe AS is defined as a mean gradient $>60\text{mmHg}$ ($V_{\text{max}} > 5\text{m/sec}$), surgery is indicated (IIA). If progression is $>0.3\text{m/sec}$ per year, surgery is indicated (IIA). AVR is reasonable when the serum B-type natriuretic peptide (BNP) level is >3 times normal (IIA).

and

Scoring Key

Score	Criteria	Synonyms
0.33	Serial echocardiogram	Follow patient with echo
0.33	Exercise test	Exercise stress test, stress test/stress echo
0.33	beta-natriuretic peptide	BNP >3 times normal
0	Listing more than 3	

Key Feature being addressed:

Given an adult presenting with severe aortic stenosis and positive exercise test, he should be considered for aortic valve replacement, either surgical or transcatheter approach. To evaluate his candidacy for TAVR, he will require CT imaging of the valve and coronary artery height. Other studies include coronary angiogram and dental evaluation.

Scoring Key

Score Correct Responses

0.33 CT Angio

0.33 Dental Evaluation

0.33 Coronary Angiogram

0 Selecting more than 3

of echocardiogram demonstrates a peak gradient of 65mm Hg and mean gradient 42 mmHg with a left ventricular ejection fraction of 65%.

What diagnostic test(s) would you order? List up to 3.

- 1: _____
- 2: _____
- 3: _____

The exercise test shows a decrease in systolic BP of 12 mmHg, what tests/evaluations would you obtain at this point? Select up to 3

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Key Features Pilot-Lessons Learned

- 2 pilots at end of QE-**optional**-provided money incentive to take seriously (\$500, \$1500 top score)
- 50 KF cases written, 26 tested, spiral exam design to maximize question exposure
- Each candidate did 6 cases, samples placed on ABTS and TSDA websites for education
- KF results scored but did NOT count for the final grade
- Psychometrics similar to QE exam with high performers on QE also scoring higher on FFE
- Correlation with QE score only 0.43-but small sample size
- KF exam items took 5" whereas MCQ QE exam items took 1" to complete
- Writing and grading KF questions harder/longer than MCQs Pilot stopped after 2 exam cycles (time intensity, candidate results similar QE v KF)
- KF concepts brought over the oral exam to emphasize critical concepts

New oral exam introduction format

Key Concepts Tested

- Recognition and stabilization of tamponade in acute Type A dissection
- Establishing cardiopulmonary bypass with true lumen cannulation (usually femoral)
- Sequence of opening the chest, cross-clamping, and initiating myocardial protection
- Cerebral protection strategy
- Decision-making for proximal and distal anastomoses

Stem

A 62-year-old man presents to the emergency department with sudden onset chest pain and hypotension. CT angiography demonstrates an acute Stanford Type A dissection with pericardial effusion consistent with tamponade. He is transferred emergently to the operating room.

Prompts

Prompt 1. On arrival in the OR with induction of anesthesia, the patient drops their BP to 50 systolic and TEE shows tamponade. What are your first priorities and how do you establish cardiopulmonary bypass?

Key: The patient contains rupture with tamponade. There are two nearly simultaneous priorities: 1) obtaining true lumen cannulation and 2) relieving the tamponade. The candidate may choose different paths—which given the challenge of the situation may be acceptable--as long as they display understanding of the two priorities. One reasonable option is to cut down on the non-dissected groin, cannulate the femoral artery and vein directly using Seldinger technique recognizing that CPB flow will be limited and then proceed with rapid opening of the chest and control of the aortic rupture. Direct aortic cannulation is not feasible in this scenario and axillary cannulation would take too long. Another option might be sternotomy, small opening in the pericardium to relieve tamponade then rapid femoral cannulation and cooling.

CHAT [ready]

You

I want a question about management of type A aortic dissection

Assistant (Claude Opus 4.6)

Great topic – type A aortic dissection management is a high-yield area for the ABTS exam. Let me search for recent evidence to identify the best knowledge gaps to test.

✓ Found 0 articles

✓ Found 10 articles

✓ Found 3 articles

Excellent – there's rich literature here. Let me search for one more specific area that's very testable.

✓ Found 0 articles

✓ Found 5 articles 

 Claude Opus 4.6  Type your message... 

QUESTION CARD

- 1** Explore
- 2 Vignette
- 3 Stem
- 4 Distractors
- 5 Review

TOPIC

Management of Acute Type A Aortic Dissection

KNOWLEDGE GAP

Three candidate gaps: (1) Malperfusion management priority – central aortic repair first vs. peripheral intervention first; (2) Extent of distal repair – hemiarch vs. total arch vs. frozen elephant trunk; (3) Arterial cannulation strategy – axillary vs. femoral vs. central aortic

EVIDENCE (18 ARTICLES)

✓ **Type-entry-malperfusion classification in a 13-year cohort of surgically treated type A acute aortic syndromes: Impact on planning and outcomes.**
Tomasi J, Escrig P, Verhoye JP – JTCVS open, 2026
PMID 42079921 

✓ **Femoral Arterial Perfusion Guided by Intraoperative Transesophageal Echocardiography for Mesenteric Malperfusion in Acute Type A Aortic Dissection.**
Ito C, Kuwahara G, Teratani H et al. – Annals of vascular diseases, 2026
PMID 41993679 

Next Question

CHAT [ready]

Extent of distal repair (hemiarch vs. total arch vs. frozen elephant trunk) – Trainees often struggle with indications for extending the repair distally. When is hemiarch sufficient, and when is total arch + frozen elephant trunk indicated (e.g., intimal tear in the arch, connective tissue disorder, malperfusion from arch branch involvement)?

Arterial cannulation strategy – Axillary artery cannulation vs. femoral cannulation vs. central aortic cannulation. Recent data (PMID 41223915) supports axillary artery as the preferred approach for antegrade cerebral perfusion and reduced malperfusion risk.

Let me push this to the question card for your review.

✓ Updated Explore in question card

The topic and candidate knowledge gaps are now on the question card.

Which knowledge gap would you like to focus on?

Malperfusion priority – central repair first vs. peripheral intervention first

Extent of distal repair – hemiarch vs. total arch vs. frozen elephant trunk

Cannulation strategy – axillary vs. femoral vs. central aortic

Or would you prefer a different angle on type A dissection management?

QUESTION CARD

- 1 Explore
- 2 Vignette
- 3 Stem
- 4 Distractors
- 5 Review

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PMID 41993679 [↗](#)

 Claude Opus 4.6 

extent of distal repair

Next Question

CHAT [ready]

Vignette (108 words):

A 52-year-old man with a history of poorly controlled hypertension presents to the emergency department with sudden-onset tearing chest pain radiating to the back. He is diaphoretic with a blood pressure of 178/92 mmHg in the right arm and 142/78 mmHg in the left arm. CT angiography demonstrates a Stanford type A aortic dissection with the **primary intimal tear located on the lesser curvature of the aortic arch**. The dissection extends into the descending thoracic aorta to the level of the diaphragm. Transthoracic echocardiography shows an ejection fraction of 60%, no pericardial effusion, and trace aortic regurgitation. Serum creatinine is 1.0 mg/dL and lactate is 1.2 mmol/L.

Key design choices:

- **Age 52** – young enough that long-term durability matters, but not so young as to suggest Marfan
- **Primary tear in the arch** – this is the critical detail that drives the correct answer (total arch replacement needed to resect the tear)
- **No malperfusion signs** (normal creatinine, normal lactate) – keeps focus on extent of repair rather than malperfusion management
- **BP differential** – realistic finding adding clinical authenticity

Please review and edit the vignette. Would you like any changes before I move to the stem?

QUESTION CARD

 Explore

 Vignette

 Stem

 Distractors

 Review

TOPIC
Management of Acute Type A Aortic Dissection

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CHAT [ready]

Proposed stems

- "What is the most appropriate surgical approach for this patient?" – Directly tests decision-making on extent of distal repair.
- "What is the optimal extent of aortic resection in this patient?" – More specifically targets the knowledge gap about how far distally to repair.
- "Which operative strategy best addresses the primary pathology in this patient?" – Emphasizes the importance of resecting the primary intimal tear.

My recommendation: Stem #2 – It most precisely tests the knowledge gap (extent of repair) and forces the examinee to consider why hemiarch alone would be insufficient when the tear is in the arch.

Correct answer: Total arch replacement with frozen elephant trunk

Rationale: When the primary intimal tear is located in the aortic arch, hemiarch replacement fails to resect the entry tear, leaving a persistent communication between true and false lumens. Total arch replacement allows complete resection of the tear. The addition of a frozen elephant trunk (antegrade stent graft into the descending aorta) promotes false lumen thrombosis in the proximal descending aorta and provides a landing zone for future endovascular intervention if needed. This is supported by multiple studies including the long-term outcomes data from Ma et al. (PMID 41614598) and the comparative data from Piccoli et al. (PMID 40985630).

✓ Updated Stem in question card



The stem and correct answer are on the card.

QUESTION CARD

 Explore

 Vignette

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 Distractors

 Review

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Next Question

CHAT [ready]

ID	Option Text	Formula	Difficulty
D1	Ascending aortic replacement with hemiarch anastomosis	Common misconception	Medium
D2	Ascending aortic replacement with separate endovascular arch stent graft	Right concept, wrong timing	Hard
D3	Total arch replacement without frozen elephant trunk	Incomplete management	Hard
D4	Medical management with blood pressure control followed by interval surgical repair	Outdated practice	Easy

Key considerations:

- D1 + D3 would create a nicely graduated question testing both "do you go beyond hemiarch?" AND "do you add FET?" – but may be very difficult
- D1 + D4 creates a broader range of difficulty, testing both the need for emergent surgery and the correct extent of repair
- D1 + D2 tests whether trainees understand why definitive open arch repair is preferred over a hybrid approach in a good-risk patient

Which 2 distractors do you want?

 Claude Opus 4.6



QUESTION CARD

 Explore

 Vignette

 Stem

 **Distractors**

 Review

TOPIC

Management of Acute Type A Aortic Dissection

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Next Question

CHAT [ready]

Vignette (108 words): A 52-year-old man with a history of poorly controlled hypertension presents to the emergency department with sudden-onset tearing chest pain radiating to the back. He is diaphoretic with a blood pressure of 178/92 mmHg in the right arm and 142/78 mmHg in the left arm. CT angiography demonstrates a Stanford type A aortic dissection with the primary intimal tear located on the lesser curvature of the aortic arch. The dissection extends into the descending thoracic aorta to the level of the diaphragm. Transthoracic echocardiography shows an ejection fraction of 60%, no pericardial effusion, and trace aortic regurgitation. Serum creatinine is 1.0 mg/dL and lactate is 1.2 mmol/L.

Stem: What is the optimal extent of aortic resection in this patient?

📄 ⬇

Option	Text
A	Ascending aortic replacement with hemiarch anastomosis
B	Total arch replacement without frozen elephant trunk
C	Total arch replacement with frozen elephant trunk

Correct Answer: C

Quality Checks:  All 6 passed

-  Vignette length (108 words) 
-  Option parallelism

QUESTION CARD

 Explore

 Vignette

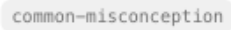
 Stem

 Distractors

 Review

CORRECT ANSWER
Total arch replacement with frozen elephant trunk

- SELECT 2 DISTRACTORS
- ☐ Ascending aortic replacement with hemiarch anastomosis 

 
- ☐ Ascending aortic replacement with separate endovascular arch stent graft 

 
- ☐ Total arch replacement without frozen elephant trunk 

 
- ☐ Medical management with blood pressure control followed by interval surgical repair 

QUESTION PREVIEW

In the Future?

AI-Powered Oral Board Training Is Available to CTSNet Users Now

August 24, 2026 |

Keywords: [Cardiac](#) [Congenital](#) [Thoracic](#)



Log in to bookmark or rate this



TAGS

Medical Technology
Training

Oral board exams assess more than clinical knowledge. They require candidates to integrate knowledge rapidly, communicate effectively, and demonstrate sound clinical reasoning in high-pressure, real-time situations. The Oral Board AI Simulator (OBAS), developed through the Ed Curtis Laboratory for Innovation in Medical Education (ECLIME) at Southern Illinois University School of Medicine, is available on CTSNet. It is designed to help cardiothoracic trainees develop and refine these essential skills.

Conclusions

- A key features exam was found to be challenging to write and grade but seemed to offer an insignificant advantage over an MCQ exam and not sufficient to replace an oral exam
- MCQ question writing shifted to clinical decision making entirely
- Starting an oral exam question with the key features helpful
- There are now many AI question writing agents and all LLM AI agents can write questions
- Surgeo is home grown and secure
- We use the term AI assisted as editing always needed
- Oasis has an AI question writing agent (Ally) that the ACC has piloted which seems encouraging

**Introducing *Resident Reach*, a new formative
assessment for residents designed by the
American Board of Anesthesiology**





Our Mission

To advance the highest standards of the practice of anesthesiology



Our Vision

To transform certification to shape the future of anesthesiology



How We Got Here & What's Ahead

- Since the 2016 introduction of MOCA Minute, programs have expressed interest in the ABA exploring a similar longitudinal assessment for residents
- In late 2024, we conducted a Resident Reach pilot with a small group of programs and about 50 residents
- The ABA's 2025 Strategic Plan solidified our strategic need to engage trainees and programs early, including a new formative assessment for residents
- The ABA has selected a specific group of programs to participate in a broader fall 2026 platform pilot of Resident Reach

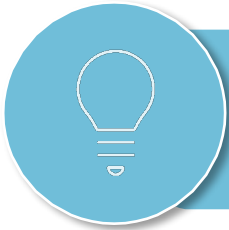




A new formative assessment for anesthesiologist trainees that aims to:



Strengthen clinical reasoning



Spark learning and reflection



Encourage dialogue between residents and teaching faculty



Offer insight into and feedback on residents' medical knowledge throughout training

What Do Residents Get Out of Resident Reach?

Residents:

- ✓ Get ongoing feedback on clinical knowledge retention throughout training
- ✓ See how their peers answered case-based questions
- ✓ Receive correct answers for cascading questions
- ✓ Answer questions weekly in ABA GO (no new platform required)



How can residents get more out of the Resident Reach pilot?

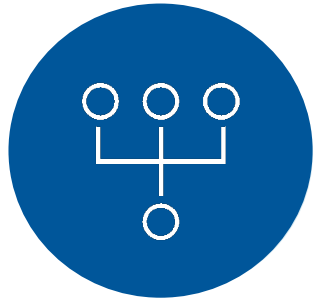
Residents should engage with attendings to discuss responses, clarify thinking and enhance understanding

What Do Programs Get Out of Resident Reach?

Program Directors:

- ✓ Gain ongoing understanding and additional insight into residents' medical knowledge throughout training, beyond ITEs
- ✓ View individual residents' live answers to case-based questions on Program Director dashboards, accessible via RTID
- ✓ Tailor conversations with residents and attendings to address gaps in knowledge
- ✓ Receive all questions on a weekly basis before residents see them for planning purposes and customized learning

Upcoming Sept. 8 Pilot Details



Participants

36 programs and **~1,500 CA-1 through CA-3 residents**, who all answer the same questions



Question Types

Include **cascading** (open-ended to multiple choice) and **case-based** (open-ended)



Not a Question Bank

Resident Reach is **not a question bank** and is intended to spark learning and discussion between residents and teaching faculty



Progress Dashboard

Program Directors can **track residents' progress** over time and identify areas where they may need additional attention

Cascading Questions

"I Know the Answer"

- Option to write in answers to questions
- Could indicate higher confidence in their clinical knowledge

"Show Me an Answer"

- Displays an answer, with option to accept or reject
- Could indicate some confidence in their clinical knowledge
- If all choices are rejected, progresses to MCQ fallback

Multiple-Choice Fallback

- Answer options presented together as a multiple-choice
- Could indicate limited confidence in their clinical knowledge
- Residents at this stage must select an option to continue

- Cover ADVANCED and BASIC Exam topics
- Residents rate their own confidence level in their responses
- After submitting their answer, residents receive the correct answer, explained with references
- Includes a points system to encourage engagement

Case-Based Questions

- New question type developed for Resident Reach with multiple questions per case (similar to SOE Additional Topic cases)
- Free-form written answers allow residents to explain clinical reasoning
- After submitting answers, residents see anonymized peer responses with no correct/incorrect feedback as there may be more than one reasonable approach
- Program Directors have immediate access to individual residents' case-based answers, which can be used to inform individual conversations
- Each question is estimated to take no more than 10 minutes

Clinical case

A 3-year-old boy with clear rhinorrhea is scheduled for adenoidectomy, bilateral myringotomy, and tubes. The patient had a productive cough 1 week ago but it has since resolved. The procedure was postponed 2 months ago for similar symptoms.

Question 1

Do you proceed as scheduled?
Defend choice.

Submit

You cannot change your answer after submission

Question 2

Assume you decide to proceed as scheduled. Is additional information necessary? If so, what additional information do you need?

My question 2 answer

Question 3

Is intubation necessary? Defend your choice.



Dashboard



Progress Report



Notifications



Menu

Case-Based Peer Responses Page

 [View Question](#)

Answers

- My answer

Question 1

would recommend proceeding. The surgery may be curative for the symptoms he is presenting with, thus if the surgery does not take place he is likely to continue developing URI symptoms and postponed further procedures. Ultimately removed the adenoids and placing tubes may help him avoid the OR in the future.

Question 2

Additional information is necessary. Any other medical history? How many other instances of rhinorrhea like symptoms has he had in the past

- Resident · CA-3

Question 1

I would proceed because the patient has mild symptoms which is associated with a lower risk for an airway emergency in a pediatric patient vs moderate or severe symptoms. However, I would still want to proceed cautiously and have a plan in place in case an airway emergency were to occur. It is possible that by delaying the procedure, the patient may return with similar symptoms; in other words, the surgery may aid in diminishing the symptoms the patient has presented with on both occasions.

- Resident · CA-2

Question 1

I would recommend proceeding. Surgery would improve symptoms and reduce risk of reoccurrence

Question 2

Recent fevers, cough, pneumonia

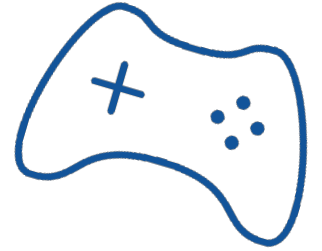
Question 3

Intubation is not necessary. Intubation irritates airways and can increase risk of bronchospasm

Question 4

Bronchospasm

Piloting a Points System

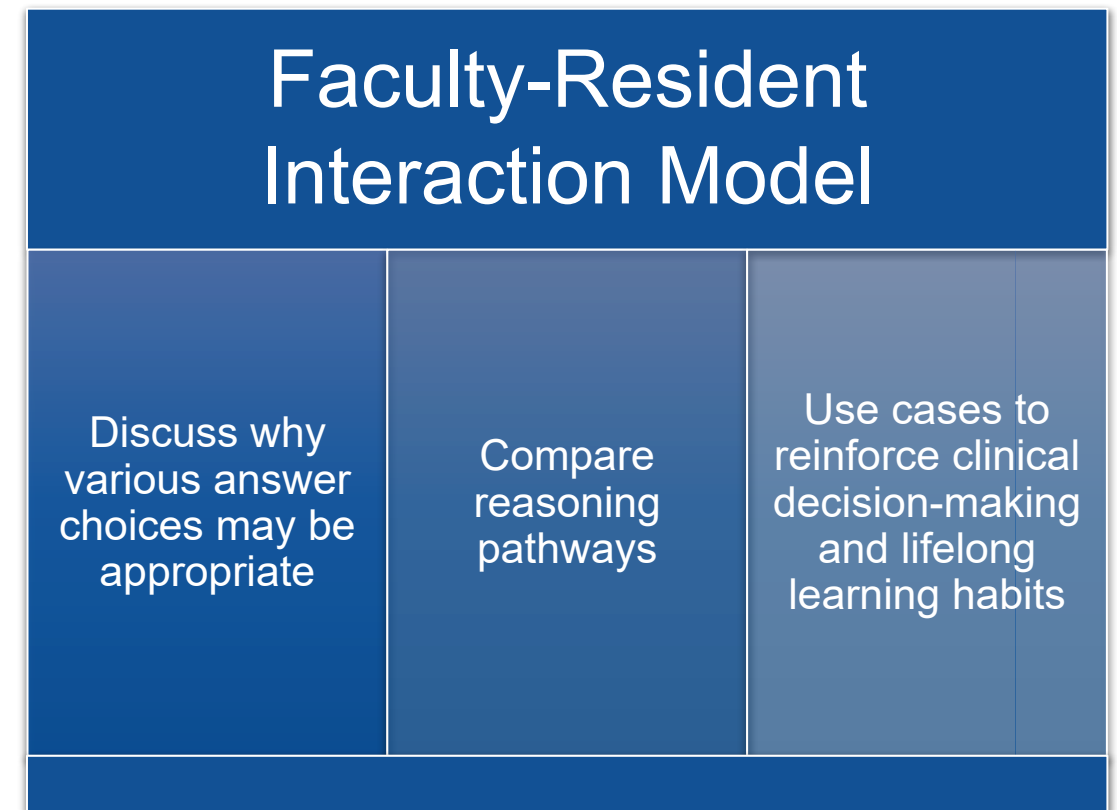


- Designed to explore residents' behavior in the pilot phase and gauge how points can incentivize participation
- Resident feedback on this may influence future iterations
- Points are awarded for answering questions to encourage engagement
- For cascading questions, more points are given for correct answers and completing more challenging question types, such as "I Know the Answer" written response
- For each clinical case, the amount of points received equates to the number of questions completed within each case
- Points are not used for evaluation, ranking or any formal assessment
- Residents can see updates on their total points in a points calculator in ABA GO; Program Directors cannot see residents' points

Engagement Suggestions

Program Directors and teaching faculty are encouraged to:

- Review weekly questions in advance; questions will be available for Program Directors to share with teaching faculty
- Use questions as teaching opportunities during educational sessions
- Promote discussion around different clinical approaches and reasoning
- Encourage participation without framing the experience as a graded assessment



What's Next?

Pilot Findings Timeline – the ABA will:





Discussion/Questions

